

Phase Transition of Traffic Overloading Failures in Large Wireless Networks

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Abstract—Traffic overloading failures are infectious due to the correlations in their occurrence. When a node fails at excessive load, the re-routed traffic may fail other nodes in the network. We present in this paper a mathematical characterization of the resilience of large wireless networks to the traffic overloading failures. We show that as the number of failed nodes increases, the network transits from a failure-resistant phase to a failure-prone phase at certain time. We derive an upper bound on this phase transition time to provide an estimation of the maximum network resilience to traffic overloading failures.

I. INTRODUCTION

Node failures pose a persistent challenge for the correct functioning of wireless networks. The challenge is intensified in large wireless networks due to the increased network complexity. With a large number of nodes, both the software and hardware requirements become complicated. Failures hence occur with an increased probability since it is difficult to keep every component in the network to work properly.

Traffic routing is one of the major network performance aspects that are under immediate impact of node failures. When a node fails, all the traversing traffic must be re-routed through alternative nodes. If the alternative nodes are not available timely, the traversing traffic cannot be delivered. Many studies have investigated the mitigation strategies to minimize the impact of node failures, such as construction of backup paths [1]–[4] and design of fast local re-routing protocols [5]–[9]. These schemes have greatly improved the routing resilience to node failures in large networks.

However, as the nodes in a network are physical devices that are designed for working under certain load limits, a node fails if its traffic load exceeds the maximum specification. Traffic re-routing effectively re-distributes the traffic load from the initial failed node to the others in the replacement paths, which possibly results in overloading failures at these nodes when they receive the extra load. Traffic re-distribution introduces causal relations among node overloading failures. The failures hence become infectious and they may propagate from one place to another in the network.

Different from the existing studies on failure countermeasures, we are interested in characterizing the failure propagation due to traffic re-distribution. We note that the occurrence

of overloading failure is determined by the sum of the existing traffic load and the extra load received from a failed node. When the existing load is low, it is unlikely for a node to fail after receiving the extra load. Nevertheless, as the number of failed nodes increases over time, each surviving node receives gradually a large total of extra load that may cause failure eventually. There thus exists a state transition of the network from the failure-resistant phase to the failure-prone phase. We characterize the time of this phase transition in this paper.

Phase transition delimits two states of the network regarding its resilience to traffic overloading failures. Before phase transition, node failures are tolerable. The re-distributed traffic from a failed node is unlikely to fail other nodes. The propagation of failures is limited. After phase transition, however, the network is in a highly unstable state. An initial failure may trigger non-stopping propagation throughout the network. We provide in this paper an analytical upper bound on the phase transition time, which serves as an estimation of the maximum network resilience to traffic overloading failures.

Specifically, we establish a mathematical model for load re-distribution and use the *percolation* theory [10], [11] to obtain the upper bound on the phase transition time. Our load re-distribution model does not specify any particular routing details and thus represents a wide class of routing protocols, which generalizes the applicability of this work to a variety of network scenarios. By using the analogy of failure propagation to liquid percolation, we determine the percentage of nodes of which the failures will drive the network from the failure-resistant phase into the failure-prone phase. Since failures occur over time, the percentage of nodes that have failed is used as an equivalent measure of time.

The rest of this paper is organized as follows. Section II presents our network model and formulates the phase transition problem. Section III determines the traffic load on an arbitrary node in the network after some percentage of nodes have failed. Section IV derives an upper bound on the percentage of failed nodes that suffices for the phase transition. Lastly, Section V concludes this paper.

II. PROBLEM MODELING AND FORMULATION

In this paper, we study the propagation of traffic overloading failures in a large wireless network. We model the network and the failure correlations as follows.

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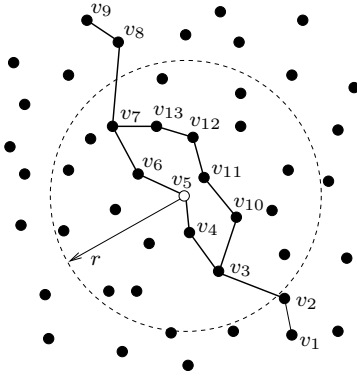


Fig. 1. Traffic re-routing and load re-distribution. Note that there are possibly a large number of traffic flows traversing node v_5 . We have only shown one of them for clarity.

A. Network Modeling

The wireless network occupies a large square area $\mathcal{B} = [-\frac{L}{2}, \frac{L}{2}]^2$ with $L \rightarrow \infty$. The nodes in the network are located randomly by following a Poisson point process with density λ . We denote the set of nodes as $\mathcal{V}$ and the number of nodes as $|\mathcal{V}| = n$, which is a random variable, and assume that all the nodes are static in this paper.

Each node communicates to a random destination. Let χ_i denote the packet rate of node v_i and $M = \{\alpha_{ij}\}$ denote the routing matrix in which $\alpha_{ij} = 1$ if the packet flow from node v_i travels through node v_j and $\alpha_{ij} = 0$ otherwise. The total traffic load on an arbitrary node v_j is then expressed as $\Gamma_j = \sum_{i=1}^n \alpha_{ij} \chi_i$. For simplicity, we assume $\chi_1 = \dots = \chi_n = \chi$. Without specifying any particular routing protocol, we simply assume that the traffic load Γ_j is a random variable and $\Gamma_1, \Gamma_2, \dots, \Gamma_n$ are distributed identically and independently. The random variable Γ_j models the initial traffic load on v_j before any occurrence of failures in the network.

Every node has a uniform and constant traffic load limit ψ ($\psi \gg \chi$), below which the node works normally and above which the node fails. When a node fails, the traversing traffic is re-routed through the other nodes to bypass the failed node, as illustrated in Fig. 1. In this example, a traffic flow travels through nodes v_1-v_9 . When v_5 has failed, the path is re-routed as $v_1-v_3, v_3-v_{10}, v_{10}-v_{13}$ and v_7-v_9 . The segment $v_{10}-v_{13}$ is a replacement for the broken segment v_4-v_6 .

Although we do not specify the particular routing protocol in this paper, we assume that it always tries to repair a broken path locally, which is studied extensively in the literature as a practical solution for dynamic routing [5]–[9]. We hence define the *path repair radius* r as the maximum distance from the site of a node failure in which other nodes can be selected to construct the replacement segment. In other words, the radius r denotes the size of the load re-distribution impact zone, as shown in Fig. 1. Note that the radius r is related closely to the node density λ . With large λ , there are many alternative nodes for selection into the replacement path and, therefore, the repair radius r is small. On the other hand, when λ is small, the radius r is correspondingly large. To reflect this inverse relation between λ and r , we require $\lambda \pi r^2 \geq \beta$ in this

paper, where β is a positive constant.

When a traffic flow is re-routed, the nodes on the broken segment (including the failed node and other working nodes bypassed, such as v_4-v_6 in Fig. 1) receive less traffic load and the nodes on the replacement segment (such as $v_{10}-v_{13}$ in Fig. 1) receive more traffic load than before, while the load on the unchanged segments (such as v_1-v_3 and v_7-v_9 in Fig. 1) remains the same as before. We use two independent probabilities p^- and p^+ ($0 < p^-, p^+ < 1$) to model the load dynamics: p^- denotes the probability for a given node (for example v_6) to reside on the original flow path prior to the failure occurrence and p^+ denotes the probability for a given node (for example v_{12}) to reside on the replacement path after failure occurrence. Thus, after the failure of a node, say v_j , and with respect to a flow χ_i that passes through v_j (implying $\alpha_{ij} = 1$), the load on an arbitrary node v_k within radius r from v_j changes in one of the three possibilities:

- i) decrease by χ_i with probability $p^-(1 - p^+)$,
- ii) increase by χ_i with probability $p^+(1 - p^-)$,
- iii) no change with probability $1 - p^- - p^+ + 2p^-p^+$.

In addition, the local repair routing protocols, e.g. [5]–[9], always find the shortest local replacement segment. As a result, as nodes fail over time and a path is re-routed repeatedly, the length of the path increases gradually, which implies that the probability for an arbitrary node v_k to be chosen into the replacement path is higher than the probability of being included in the previous path. We thus assume $p^+ > p^-$ in order to be consistent with this observation. Besides, with respect to all the nodes within radius r , since a particular flow traverses comparatively a small subset of these nodes, we assume that p^- and p^+ are small positive constants.

We focus on the traffic overloading failures in this paper and assume that all the initial random failures are also caused by overloading. Since Γ_j is a random variable, it may exceed the load limit ψ at some time when the random traffic flows change in the network. In addition, we assume that different nodes are selected into flow paths independently and different flows are routed independently.

B. Problem Formulation

Using the above network model, we are able to mathematically describe the correlations among traffic overloading failures. When a node v_j fails, all the traversing traffic flows are re-routed away from it, resulting in the traffic load changes in other nodes. For each flow χ_i that traverses v_j , an arbitrary node v_k within radius r from v_j decreases its load by amount χ_i with probability p^- if χ_i flows through v_k before the failure of v_j , and v_k increases its load by amount χ_i with probability p^+ if χ_i traverses v_k after re-routing. The load increment and decrement are independent from each other. Since $p^+ > p^-$, the load on v_k is expected to increase. Let $\Delta_k(v_j)$ denote the load change on v_k after the failure of v_j . v_k fails if $\Gamma_k + \Delta_k(v_j) > \psi$ and does not fail if $\Gamma_k + \Delta_k(v_j) \leq \psi$.

When the initial traffic load in the network is low, the re-distributed load received from v_j is unlikely to fail v_k since v_k has plenty of unused capacity. As the number of failed nodes

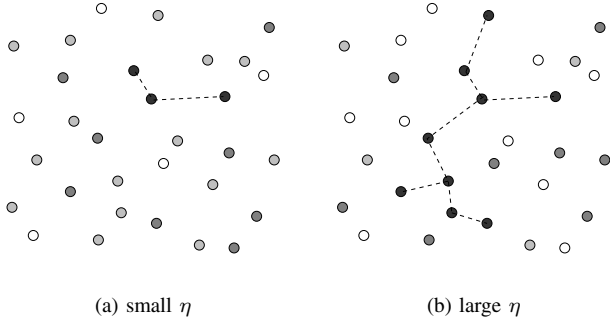


Fig. 2. An example of load dynamics and failure propagation. The white nodes have failed and the darkness on the other nodes is proportional to the traffic load. In the failure-resistant phase (a), η is small and the loads are light in the network. When failure occurs, its propagation is limited, as shown by the small cluster of darkest nodes connected by the dash lines. In the failure-prone phase (b), η is large and the loads become heavy in the network. When failure occurs, its propagation is unbounded, as shown by the large cluster of darkest nodes connected by the dash lines.

increases over time, the capacity of v_k is filled up gradually such that $\Gamma_k + \Delta_k(\tilde{\mathcal{V}})$ approaches ψ and v_k becomes unstable, where $\Delta_k(\tilde{\mathcal{V}})$ denotes the total received load at v_k after a subset $\tilde{\mathcal{V}}$ of nodes have failed. Node v_k fails eventually when $\Gamma_k + \Delta_k(\tilde{\mathcal{V}}) > \psi$. When v_k fails, its total load is re-distributed to other nodes again via re-routing.

We define the *failure percentage* $\eta = \frac{|\tilde{\mathcal{V}}|}{|\mathcal{V}|}$ as a time measure for the network evolution with node failures. With η increasing over time, the loads on the surviving nodes increase and the network transits from the failure-resistant phase into the failure-prone phase. For rigorous presentation, we define the concepts of these two phases below.

Definition 1: Failure cluster. A group of failed nodes form a failure cluster $\mathcal{C}(v_j)$ if all the failures (except v_j) in this cluster are results of receiving the re-distributed load from v_j or from any node that fails after v_j . The failure cluster size is denoted as $|\mathcal{C}(v_j)|$.

Definition 2: Failure-resistant phase. The network is in the failure-resistant phase if $\forall v_j, \Pr[|\mathcal{C}(v_j)| = \infty] = 0$.

Definition 3: Failure-prone phase. The network is in the failure-prone phase if $\forall v_j, \Pr[|\mathcal{C}(v_j)| = \infty] > 0$.

Definition 4: Critical failure percentage. The critical failure percentage is defined as

$$\eta_c = \inf\{\eta > 0 : \Pr[|\mathcal{C}(v_j)| = \infty] > 0\}. \quad (1)$$

In the term of percolation, we say that failure does not percolate if the network is in the failure-resistant phase and failure percolates if the network is in the failure-prone phase. An example of the network load dynamics and failure propagation is illustrated in Fig. 2.

The critical failure percentage η_c delimits the two network phases. Our objective is to derive an upper bound $\bar{\eta}_c$ on η_c such that the network must have entered the failure-prone phase by the time of $\bar{\eta}_c$, which provides an estimation of the maximum network resilience to the traffic overloading failures. Obviously, η_c may not exist if λ is extremely small, in which the total re-distributed load received by an arbitrary node v_k

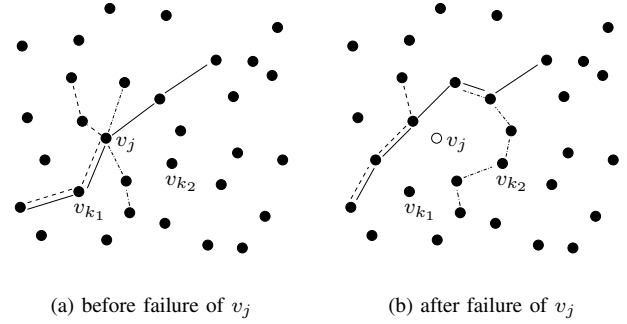


Fig. 3. An example of load changes after the node failure at v_j . Before the failure, three traffic flows pass through v_j , as indicated by the three lines of different styles in (a). After the failure, all the three paths are re-routed as shown in (b). As the result of re-routing, the load decreases on v_{k1} by amount 2χ and increases on v_{k2} by amount χ .

is not sufficient to fill up its capacity even if all the nodes within radius r of v_k have failed. In order to proceed with a meaningful study, we assume the existence of a small positive constant ϵ such that failure percolates when $\eta = 1 - \epsilon$.

In the next, we study the phase transition of traffic overloading failures in two steps. We first determine the traffic load on an arbitrary node after η percentage of nodes have failed and then derive an upper bound on $\bar{\eta}_c$ for phase transition.

III. DETERMINATION OF LOAD DYNAMICS

We now characterize the traffic load on an arbitrary surviving node when failures happen randomly over time. Because we assume that all the failures in the network are due to overloading, when a node v_j fails, its total traffic load must equal ψ , which is the summation of all the traversing flows before the failure.

Given a flow χ_i and an arbitrary node v_k within radius r from v_j , the load on v_k decreases by χ_i with probability p^- and increases by χ_i with probability p^+ . Since all the traffic flows on v_j must be re-routed, v_k has a chance to change its load in accordance to each flow χ_i . An example of the load changes on v_k after the failure of v_j is given in Fig. 3.

Let random variables $Z_k^-(v_j)$ and $Z_k^+(v_j)$ denote the total number of times that v_k decreases and increases its load after the failure of v_j , respectively. Due to the independence of flow re-routings from one to another, $Z_k^-(v_j)$ and $Z_k^+(v_j)$ follow binomial distributions: $Z_k^-(v_j) \sim B(\frac{\psi}{\chi}, p^-)$ and $Z_k^+(v_j) \sim B(\frac{\psi}{\chi}, p^+)$. Given that p^- and p^+ are small positive constants and $\psi \gg \chi$, as assumed in the network model, we can approximate the binomial distributions of $Z_k^-(v_j)$ and $Z_k^+(v_j)$ by using Poisson distributions, which are easier for analysis. Therefore, we have $Z_k^-(v_j) \sim \text{Pois}(\frac{\psi p^-}{\chi})$ and $Z_k^+(v_j) \sim \text{Pois}(\frac{\psi p^+}{\chi})$. The accumulated decreased and increased loads on v_k are hence also approximated by Poisson distributions as $\Delta_k^-(v_j) \sim \text{Pois}(\psi p^-)$ and $\Delta_k^+(v_j) \sim \text{Pois}(\psi p^+)$ respectively, and the total changed load is $\Delta_k(v_j) \sim \text{Pois}(\psi(p^+ - p^-))$.

The analysis above demonstrates that the load received by v_k after the failure of v_j is approximately a Poisson random variable with parameter $\psi(p^+ - p^-)$. As nodes fail over time,

v_k receives gradually a large amount of re-routed traffic from different failed nodes. By the property of Poisson distribution, the total received load $\Delta_k(\tilde{\mathcal{V}})$ is also a Poisson random variable, the parameter of which is $|\tilde{\mathcal{V}}_k|\psi(p^+ - p^-)$, where $|\tilde{\mathcal{V}}_k|$ is the number of failed nodes that may change the load on v_k . We next determine the variable $|\tilde{\mathcal{V}}_k|$.

Given the path repair radius r , any failure that occurs within radius r from v_k may change the load on v_k . Thus, all the nodes that fail within a circular area of πr^2 centered at v_k should be counted into $|\tilde{\mathcal{V}}_k|$. By the definition of Poisson point process, the number of nodes that fall into the circular area is a Poisson random variable with parameter $\lambda\pi r^2$. When η percentage of nodes have failed, as they occur randomly in the network, the number of failed nodes $|\tilde{\mathcal{V}}_k|$ in the circular area is a Poisson random variable with parameter $\eta\lambda\pi r^2$.

However, the randomness of $|\tilde{\mathcal{V}}_k|$ renders our analysis of $\Delta_k(\tilde{\mathcal{V}})$ difficult. To proceed with our study, we next provide a probabilistic bound on $|\tilde{\mathcal{V}}_k|$. From the Chebyshev's inequality [12], a random variable X has the following property

$$\Pr[|X - E(X)| \geq a] \leq \frac{\text{Var}(X)}{a^2}, \quad (2)$$

where a is a positive constant. Substituting $|\tilde{\mathcal{V}}_k|$ for X , $\eta\lambda\pi r^2$ for $E(X)$, $\eta\lambda\pi r^2$ for $\text{Var}(X)$ and $\frac{1}{2}\eta\lambda\pi r^2$ for a , we have

$$\Pr[||\tilde{\mathcal{V}}_k| - \eta\lambda\pi r^2| \geq \frac{1}{2}\eta\lambda\pi r^2] \leq \frac{\eta\lambda\pi r^2}{(\frac{1}{2}\eta\lambda\pi r^2)^2}, \quad (3)$$

from which we further have

$$\begin{aligned} \Pr[||\tilde{\mathcal{V}}_k| - \eta\lambda\pi r^2| \leq \frac{1}{2}\eta\lambda\pi r^2] &\geq 1 - \frac{\eta\lambda\pi r^2}{(\frac{1}{2}\eta\lambda\pi r^2)^2} \\ &= 1 - \frac{4}{\eta\lambda\pi r^2} \\ &\geq 1 - \frac{4}{\eta\beta}. \end{aligned} \quad (4)$$

Note that in order to have $1 - \frac{4}{\eta\beta} > 0$, we require $\eta > \eta_1 = \frac{4}{\beta}$.

Eq. (4) demonstrates that, with at least probability $1 - \frac{4}{\eta\beta}$, $\frac{1}{2}\eta\lambda\pi r^2 \leq |\tilde{\mathcal{V}}_k| \leq \frac{3}{2}\eta\lambda\pi r^2$ and consequently $\Delta_k(\tilde{\mathcal{V}}) \geq \Delta'_k(\tilde{\mathcal{V}})$, where $\Delta'_k(\tilde{\mathcal{V}}) \sim \text{Pois}(\frac{1}{2}\eta\lambda\pi r^2\psi(p^+ - p^-))$. Furthermore, since $\lambda\pi r^2 \geq \beta$ and we require $\eta > \frac{4}{\beta}$, we have $\eta\lambda\pi r^2 \geq \eta\beta > 4$ and $\Delta'_k(\tilde{\mathcal{V}}) > \Delta''_k(\tilde{\mathcal{V}})$, where $\Delta''_k(\tilde{\mathcal{V}}) \sim \text{Pois}(2\psi(p^+ - p^-))$. Therefore, we can characterize the load on an arbitrary node v_k in the network after η percentage of nodes have failed as a random variable $T_k(\eta)$ such that with at least probability $1 - \frac{4}{\eta\beta}$

$$T_k(\eta) \geq \Gamma_k + \Delta'_k(\tilde{\mathcal{V}}) \geq \Delta'_k(\tilde{\mathcal{V}}) > \Delta''_k(\tilde{\mathcal{V}}). \quad (5)$$

Our following derivation of the upper bound on the phase transition time is based on Eq. (5).

IV. UPPER BOUND ON THE PHASE TRANSITION TIME

As the number of failed nodes increases, the traffic load on each surviving node also increases, as seen in the random variable $\Delta'_k(\tilde{\mathcal{V}}) \sim \text{Pois}(\frac{1}{2}\eta\lambda\pi r^2\psi(p^+ - p^-))$: when η increases, $\Delta'_k(\tilde{\mathcal{V}})$ is expected to increase. Therefore, when η

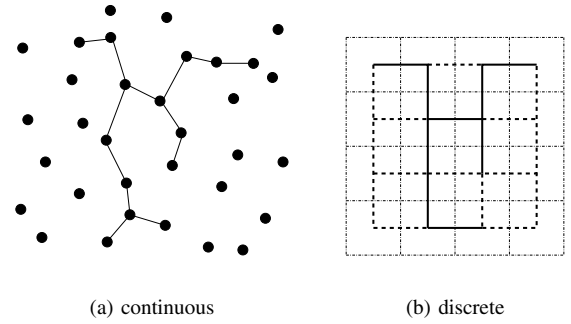


Fig. 4. Mapping from continuous percolation to discrete percolation. In (a), the nodes are randomly located. In (b), the solid lines represent open bonds and the dash lines represent closed bonds.

is large enough, one more failure in the network may start a non-stopping sequence of failure propagation. We use the percolation theory to study the failure propagation.

The percolation theory [10], [11] originated from the phenomenon of liquid transfusion in a solid and porous object. The solid object contains many randomly distributed tiny interstices and with certain probability neighboring interstices are connected by open paths that allow the liquid to pass through. When the object is wetted with the liquid at one of its interstices, percolation theory tells us whether the liquid transfusion is confined in a limited space or spread to the entire object, as determined by the probability of existing open paths between neighboring interstices.

We next determine the upper bound on the phase transition time by using the percolation theory. Fig. 4(a) illustrates an example of failure percolation. As the nodes are randomly located, the percolation model in Fig. 4(a) is known as *continuous percolation*. Because the randomness of node locations complicates the derivation, we first map the continuous percolation into a *discrete percolation* problem, which is easier for our analysis.

As shown in Fig. 4(b), we divide the network into many small square cells and represent each cell as a site. Two neighboring sites are connected by a bond, which is either *open* or *closed*. The bond is open if failure propagates from one site to the other, and closed if failure does not propagate between the two connected sites. Let p denote the probability that a bond is open. Known from the theory of discrete percolation [10], percolation happens if $p > 0.5$ and does not happen if $p < 0.5$. Therefore, our task is to determine the percentage of failed nodes η at which $p > 0.5$ is reached.

Note that a site in the discrete percolation represents a cell. Hence, a bond connecting two sites represents the aggregation of all the failure propagations between two neighboring cells. As long as failure propagates from any node in one cell to any node in the other cell, the bond connecting these two cells is open. Keeping this representation in mind, we choose the edge length of each cell to be $\frac{r}{\sqrt{5}}$ such that any two nodes in two neighboring cells are within distance r from each other, as shown in Fig. 5(a). With this small cell size, however, failure may propagate in one hop from a cell to non-neighboring cells,

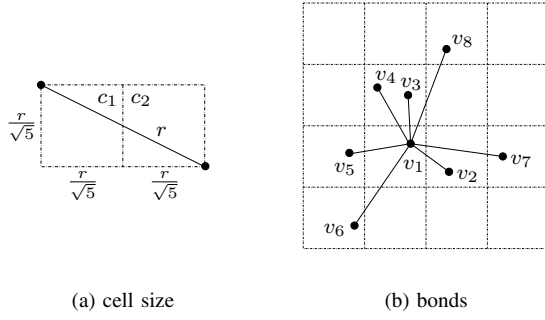


Fig. 5. Determination of the cell size and example of the bonds between neighboring cells and non-neighboring cells. When the edge of each cell is $\frac{r}{\sqrt{5}}$, failure may propagate from a node in cell c_1 to a node in cell c_2 , regardless of the specific locations of these two nodes in their respective cells. With this small cell size, open bonds may appear between neighboring cells (such as $v_1-v_2, v_1-v_3, v_1-v_4, v_1-v_5$) and between non-neighboring cells (such as $v_1-v_6, v_1-v_7, v_1-v_8$).

as illustrated in Fig. 5(b). We note that the open bonds between non-neighboring cells can be ignored safely, because they do not change the validity of our result in this paper. When η is large enough such that failure percolates using only the bonds between neighboring cells, it also percolates by using the bonds between non-neighboring cells in addition to those between neighboring cells.

Next, we derive the probability p for a bond to be open. Given a failed node v_j and another node v_k in a neighboring cell, the load on v_k increases after the failure of v_j by a Poisson random variable $\Delta_k(v_j)$ with parameter $\psi(p^+ - p^-)$. Assuming that η percentage of nodes have failed prior to v_j , the load on v_k after the failure of v_j is then $T_k(\eta, v_j) = T_k(\eta) + \Delta_k(v_j)$. By Eq. (5), with at least probability $1 - \frac{4}{\eta\beta}$, we have

$$T_k(\eta, v_j) > \Delta_k''(\tilde{\mathcal{V}}) + \Delta_k(v_j) = T_k'(\eta, v_j), \quad (6)$$

where $T_k'(\eta, v_j) \sim \text{Pois}(3\psi(p^+ - p^-))$. By defining $p(v_j \rightarrow v_k)$ as the probability that v_k fails after v_j , with at least probability $1 - \frac{4}{\eta\beta}$, we have

$$\begin{aligned} p(v_j \rightarrow v_k) &= \Pr[T_k(\eta, v_j) > \psi] \\ &> \Pr[T_k'(\eta, v_j) > \psi] \\ &= \sum_{m=\psi+1}^{\infty} e^{-3\psi(p^+ - p^-)} \frac{(3\psi(p^+ - p^-))^m}{m!} \end{aligned} \quad (7)$$

We rewrite Eq. (7) as $p(v_j \rightarrow v_k) > q$ by defining $q = \sum_{m=\psi+1}^{\infty} e^{-3\psi(p^+ - p^-)} \frac{(3\psi(p^+ - p^-))^m}{m!}$. Note that the bond between two neighboring cells is open as long as failure propagates from any node in one cell to any node in the other cell. We thus obtain

$$\begin{aligned} p &\geq 1 - e^{-(1-\eta)\lambda(\frac{r}{\sqrt{5}})^2 p(v_j \rightarrow v_k)} \\ &\geq 1 - e^{-\frac{\epsilon\beta}{5\pi} p(v_j \rightarrow v_k)} \\ &> 1 - e^{-\frac{\epsilon\beta q}{5\pi} (1 - \frac{4}{\eta\beta})}, \end{aligned} \quad (8)$$

because the number of surviving nodes in a cell is Poisson distributed with density $(1 - \eta)\lambda$ and each of them fails with

probability $p(v_j \rightarrow v_k)$ after the failure of v_j . Letting

$$1 - e^{-\frac{\epsilon\beta q}{5\pi} (1 - \frac{4}{\eta\beta})} > 0.5, \quad (9)$$

we finally arrive at

$$\eta > \eta_2 = \frac{4\epsilon q}{\epsilon\beta q - 3.4657\pi}. \quad (10)$$

Combining Eq. (10) and the requirement $\eta > \eta_1 = \frac{4}{\beta}$, we have thus obtained a sufficiently large η at which failure percolates in the network. This large η also serves as the upper bound on the phase transition time such that $\bar{\eta}_c = \max\{\eta_1, \eta_2\}$.

V. CONCLUSION

In this paper, we have studied the resilience of large wireless networks to the traffic overloading failures. As traffic changes randomly in the network, some nodes may fail when their routing loads exceed the designed maximum specification. Their failures increase the loads on the surviving nodes and possibly result in further failures. We have mathematically modeled the spreading of traffic overloading failures and characterized the network resilience by using two phases: failure-resistant and failure-prone. Using the percolation theory, we have determined an upper bound on the time of network transition from the failure-resistant phase into the failure-prone phase, in terms of the percentage of failed nodes in the network. The work in this paper provides an estimation of the maximum resilience of large wireless networks to the traffic overloading failures.

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